

Human Computer Interaction (HCI)

Usable User interfaces



Mah Noor Fatima
CS Lecturer
CUI Sahiwal Campus

What is Usability?



“Extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.”
- ISO 9241-11

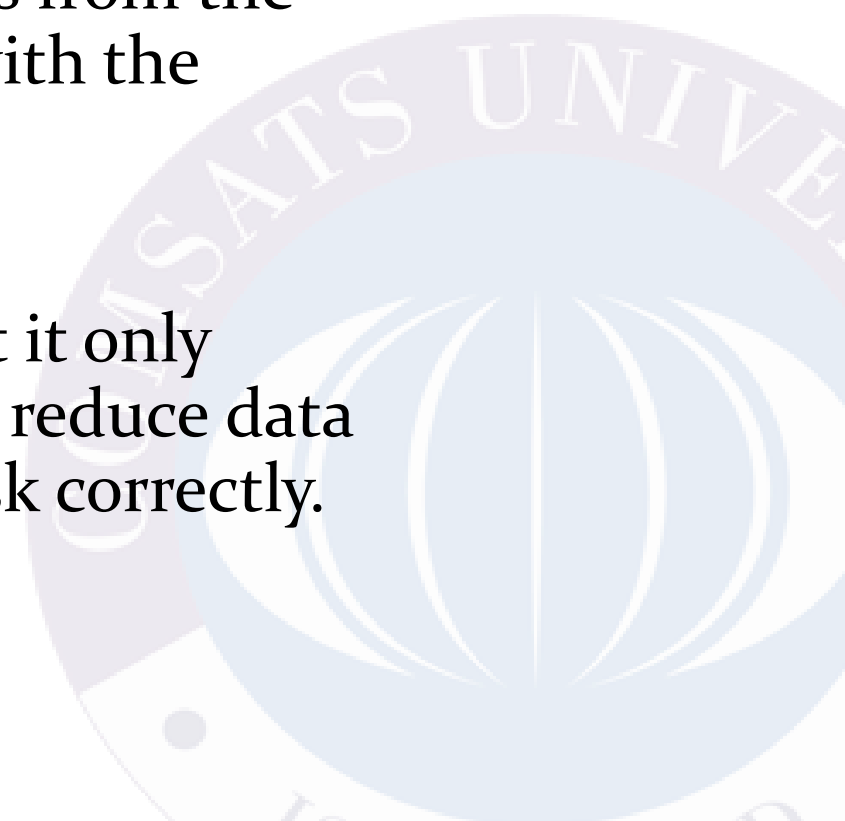


Characteristics of Usable product

- 1) Effectiveness: Effectiveness is about whether users can complete their goals with a high degree of accuracy.

Much of the effectiveness of a product comes from the support provided to users when they work with the product.

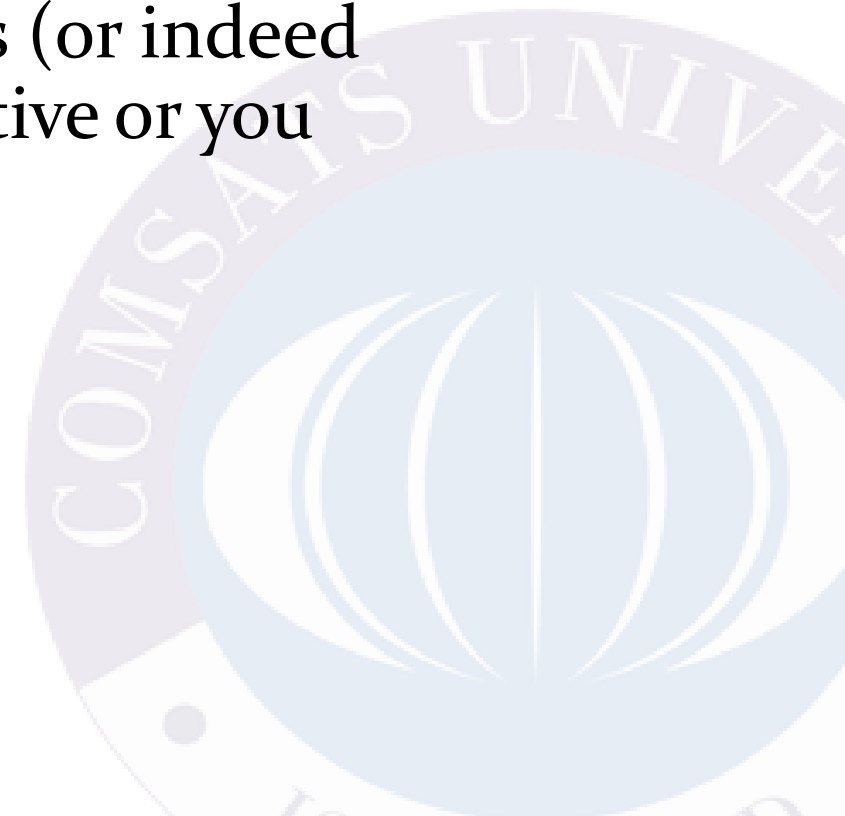
For example, fixing a credit card field so that it only accepts a valid credit card number entry can reduce data entry errors and help users perform their task correctly.



Characteristics of Usable products

2) Efficiency: Efficiency is all about speed. How fast can the user get the job done?

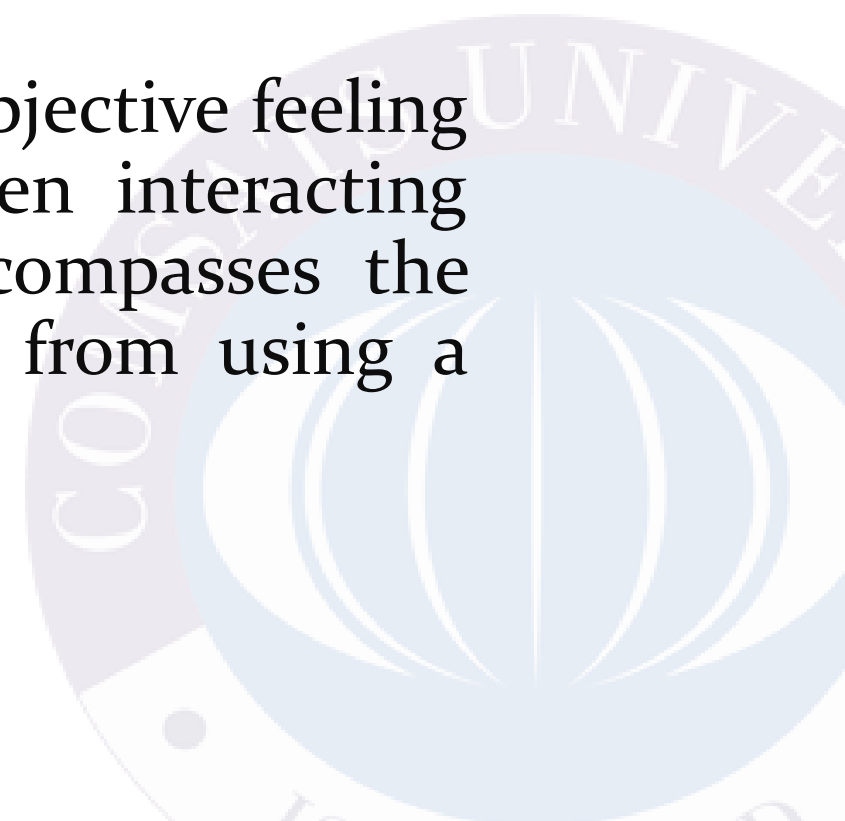
You need to examine the number of steps (or indeed clicks/keystrokes) to achieving the objective or you need to record the time.



Characteristics of Usable product

3) Satisfaction: Freedom from discomfort and positive attitudes towards the use of the product.

-Satisfaction in usability refers to the user's subjective feeling of contentment, pleasure, or fulfillment when interacting with a product, system, or interface. It encompasses the overall positive experience that users derive from using a product or completing a task within a system.



Key factors that contribute to satisfaction in usability include:

- **Ease of Use**

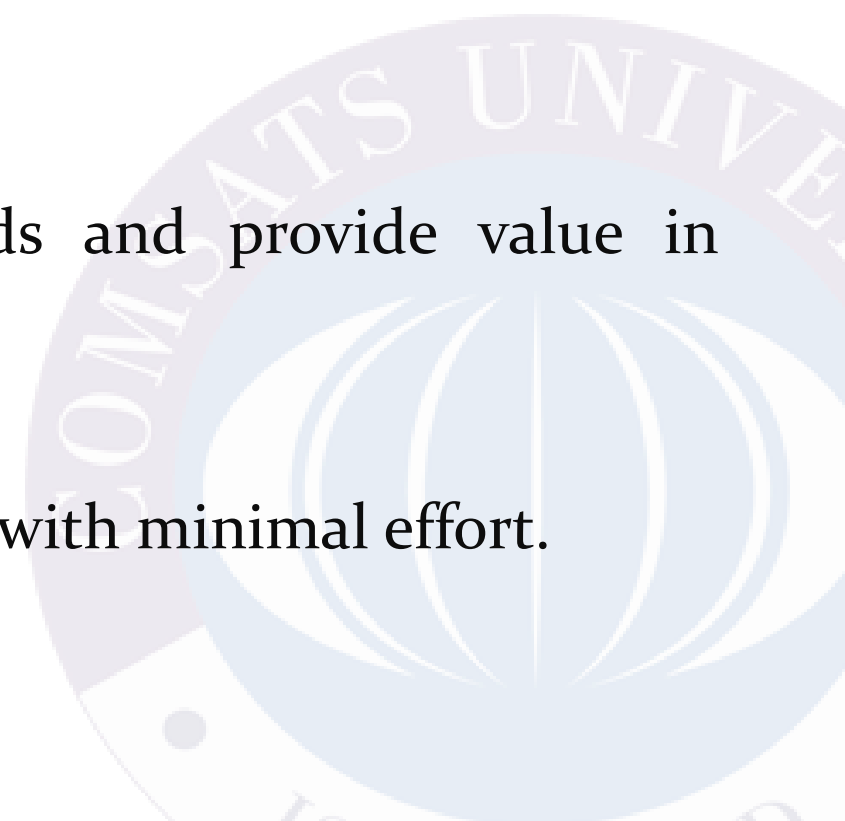
Users should find the product or system easy to navigate, understand, and interact with

- **Usefulness**

The product or system should fulfill users' needs and provide value in accomplishing their goals or tasks.

- **Efficiency**

Users should be able to accomplish tasks quickly and with minimal effort.



- **Error Tolerance**

The product or system should be forgiving of user errors and provide clear feedback and guidance to help users recover from mistakes. Users are more likely to feel satisfied if they can easily correct errors and continue their tasks without frustration.

- **Aesthetics**

The visual design of the product or system plays a significant role in user satisfaction. A visually appealing interface that employs pleasing colors, typography, and layout contributes to a positive user experience.



Nielsen and Molich's 10 User Interface Design Guidelines



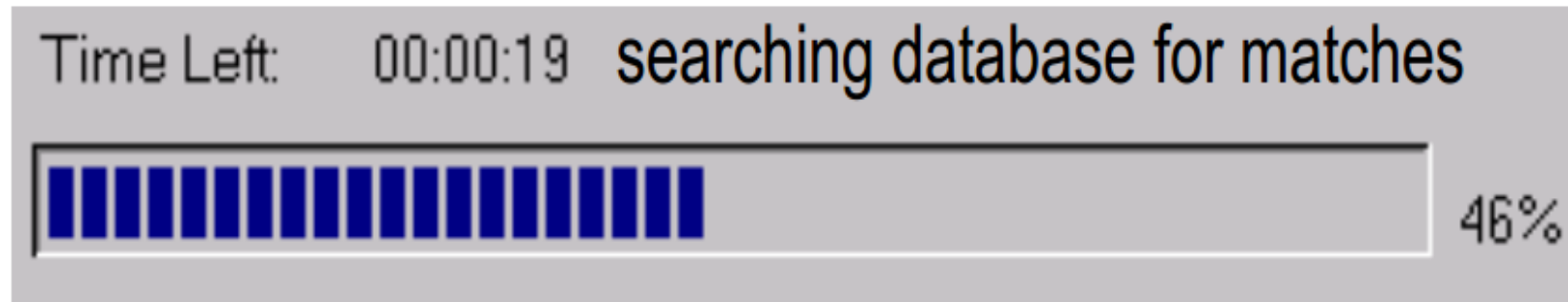
Nielsen and Molich's 10 User Interface Design Guidelines

- Jakob Nielsen, a renowned web usability consultant and partner in the Nielsen Norman Group, and Rolf Molich, another prominent usability expert, established a list of ten user interface design guidelines in the 1990.
- These heuristics have been used in many of the products designed by some of the most successful companies in the world such as Apple, Google, and Adobe.

Nielsen and Molich's 10 User Interface Design Guidelines

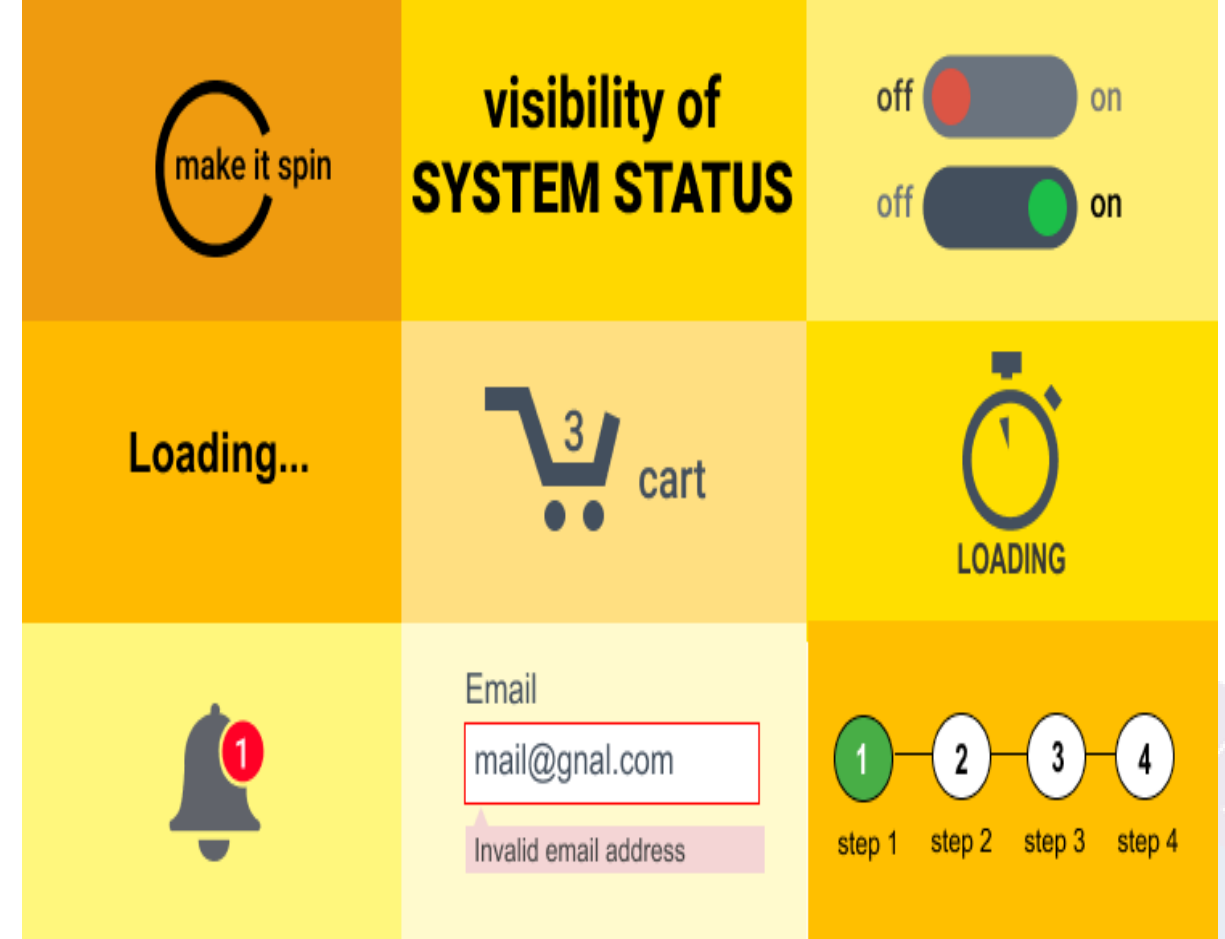
1) Visibility of system status:

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.



Importance

- When users know the current system status, they learn the outcome of their prior interactions and determine next steps.
- Predictable interactions create trust in the product as well as the brand. **A lack of information often equates to a lack of control.**



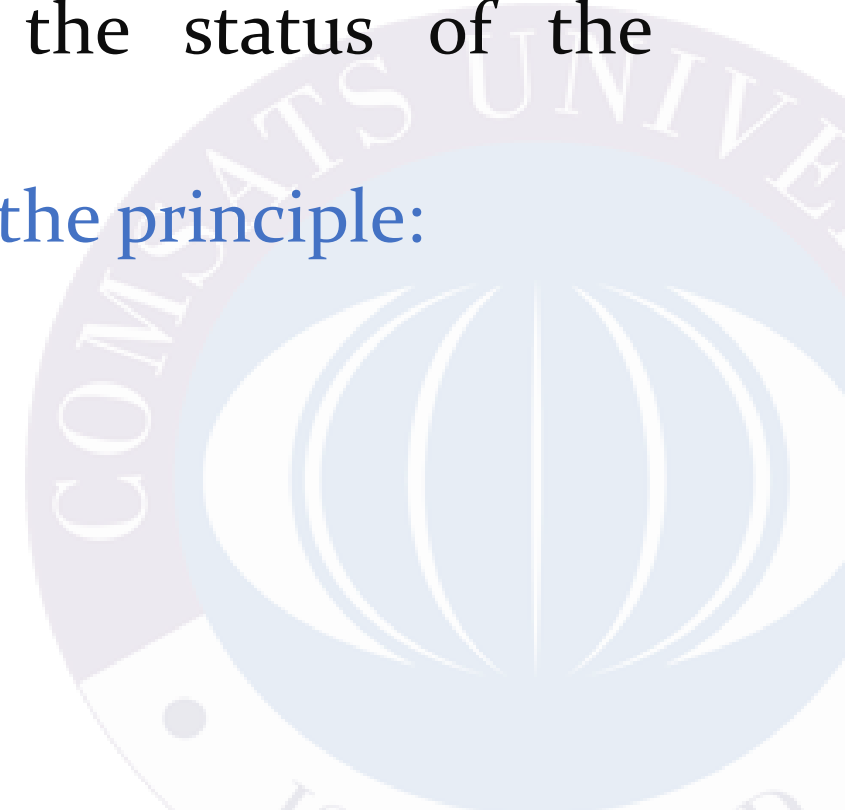
For example, your phone or laptop display how much battery life remains; your email application tells you how many unread emails you have; a sign in the subway indicates the next stop (or, if you've missed the last one, how many minutes until the next train arrives at the station) **Imagine the life without these**

Example

- An example of an interface that effectively utilizes the principle of "Visibility of System Status" is the file download interface in web browsers.
- When users initiate a file download, the browser typically provides immediate feedback to indicate the status of the download process.

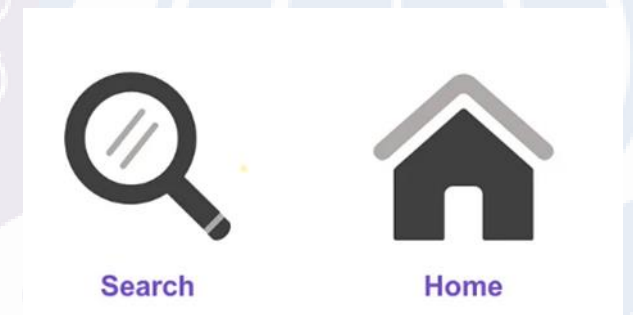
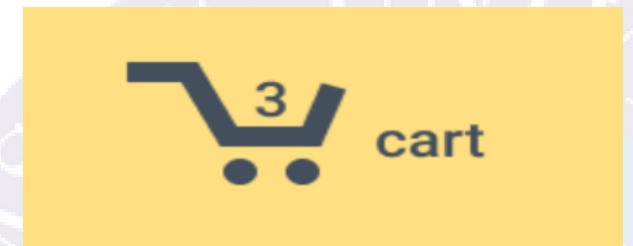
Here's how this interface example aligns with the principle:

- **Download Progress Indicator**
- **Download Speed and Time Remaining**
- **Download Confirmation**
- **Error Messages**



Nielsen and Molich's 10 User Interface Design Guidelines

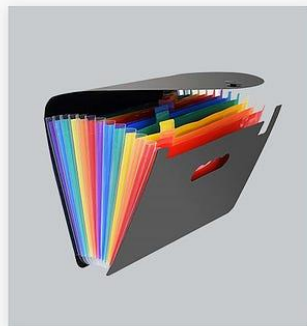
2) **Match between system and real world:** The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.



Importance

- When a design's controls follow real-world conventions and correspond to desired outcomes (called natural mapping), it's easier for users to learn and remember how the interface works. This helps to build an experience that feels intuitive.

Real World



System



example of an interface that effectively employs the principle of "Match Between System and the Real World" is the interface of a navigation app such as Google Maps

- **Map Representation:** Navigation apps display maps that closely resemble real-world geographic features, including roads, landmarks, and points of interest.
- **Symbols and Icons:** The apps use familiar symbols and icons to represent various elements on the map, such as roads, intersections, traffic lights, and destinations. These symbols are consistent with real-world signage and conventions, making them easy for users to understand and navigate.

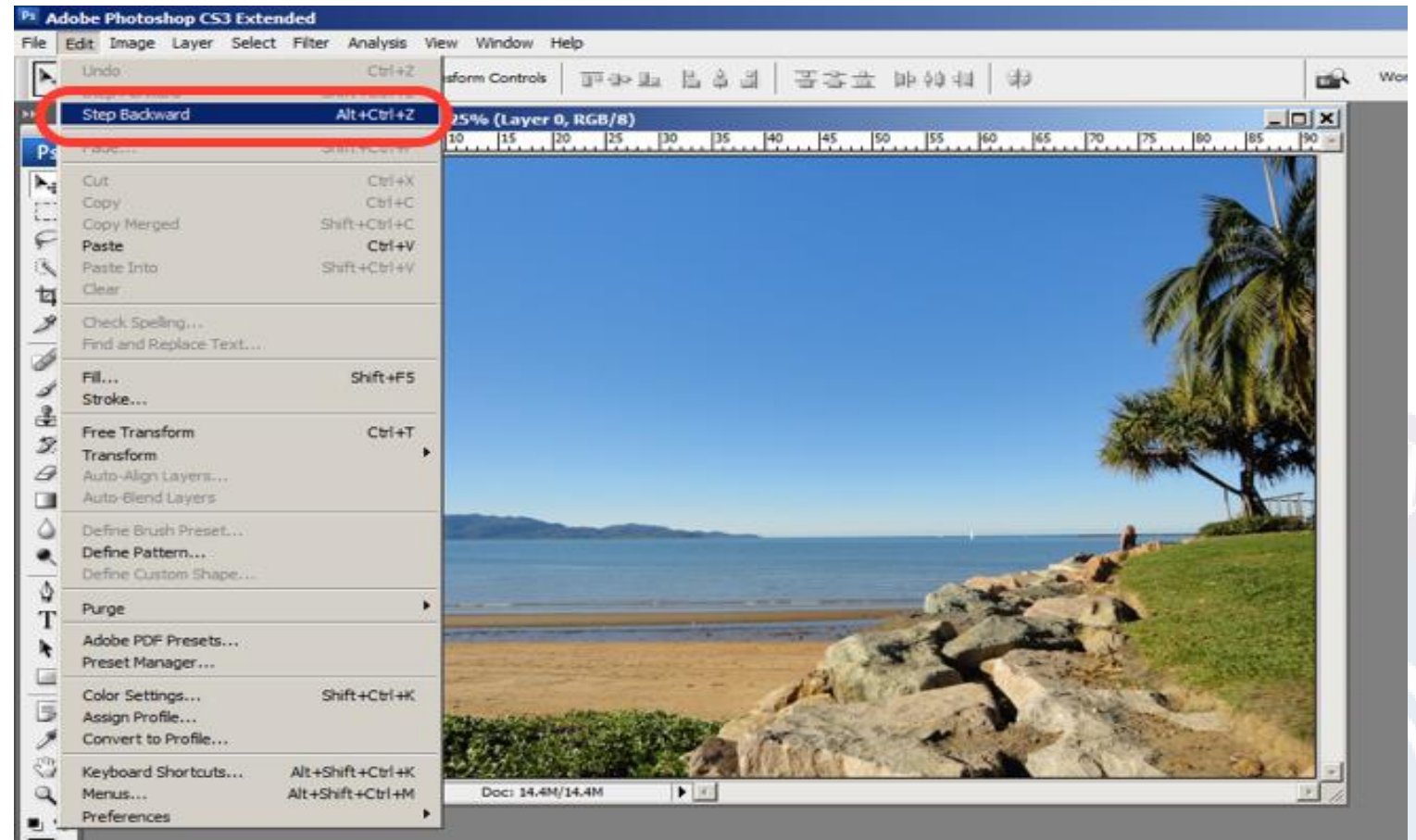
- **Directional Instructions:** Navigation apps provide users with turn-by-turn directions and guidance to help them reach their destinations. The instructions are presented in a language and format that matches real-world driving or walking directions, such as "turn left at the next intersection" or "continue straight for 500 meters."
- **Voice Guidance:** Many navigation apps offer voice-guided navigation, where users receive spoken instructions and prompts as they navigate their routes. The voice instructions mimic the way a passenger or navigator would provide directions in real-world situations, enhancing the naturalness and familiarity of the experience.

Nielsen and Molich's 10 User Interface Design Guidelines...

3) **User control and freedom:** Users often perform actions by mistake. Support undo and redo, as well as clear exit strategies, to allow users to easily recover from errors.



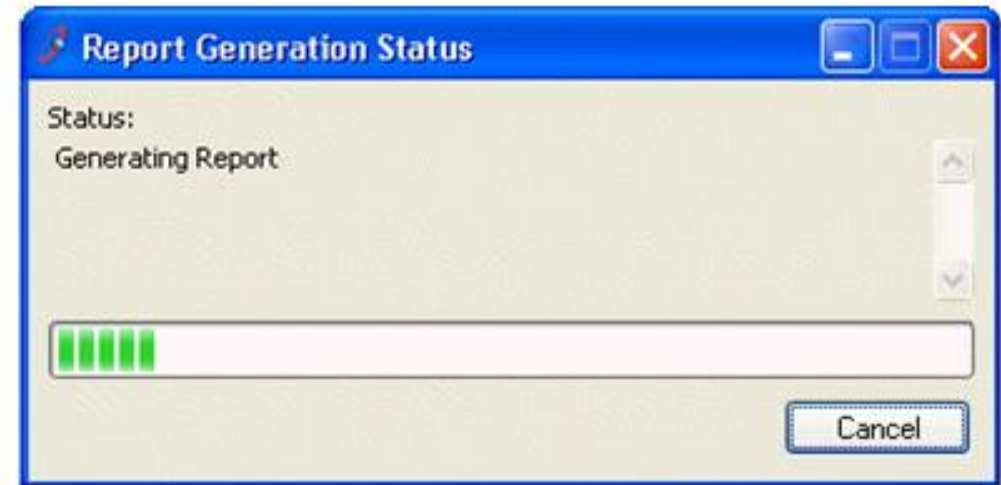
Nielsen and Molich's 10 User Interface Design Guidelines...



Example: In Photoshop when user makes changes to an image or adds various artistic effects, they are able to quickly and easily take a step backwards if they make an error.

Importance

- When it's easy for people to back out of a process or undo an action, it fosters a sense of freedom and confidence. Exits allow users to remain in control of the system and avoid getting stuck and feeling frustrated
 - Support Undo and Redo.
 - Show a clear way to exit the current interaction, like a Cancel button.



Let's apply the "User Control and Freedom" heuristic to WhatsApp.

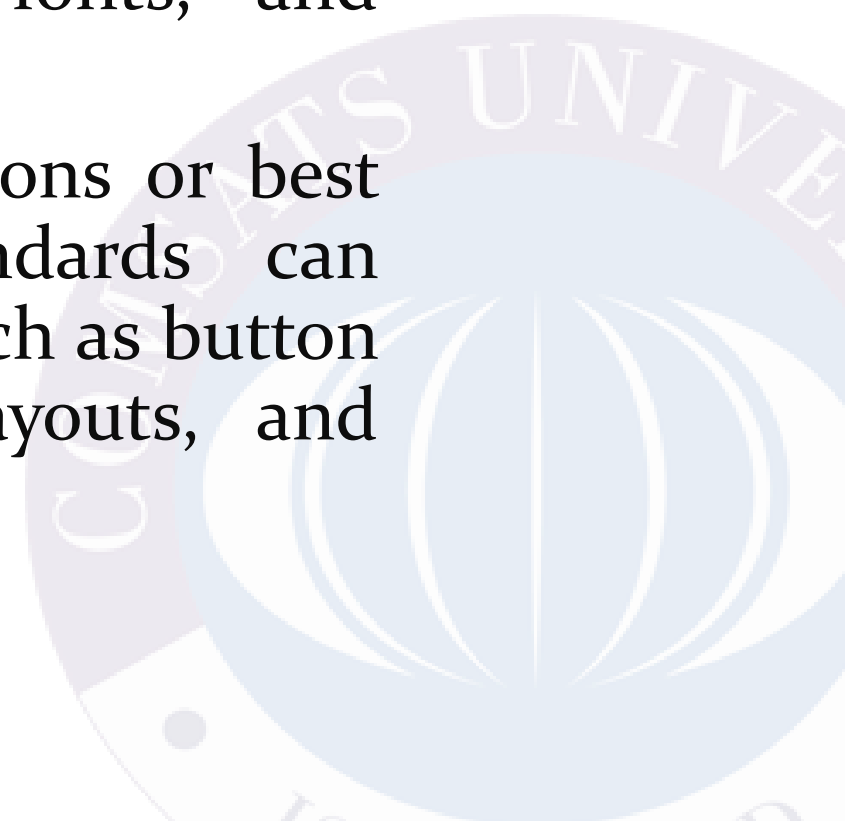
- **Message Deletion**
- **Privacy Settings**
- **Mute and Archive Chats**
- **Message Reply and Forwarding**
- **Account Settings**



Nielsen and Molich's 10 User Interface Design Guidelines...

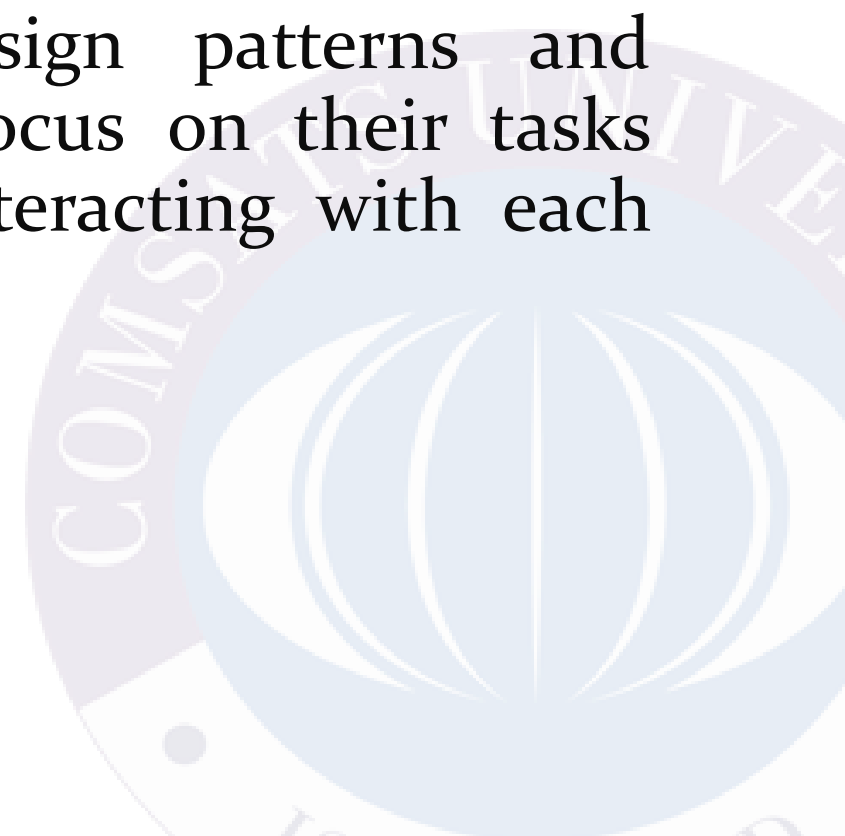
4) **Consistency and standards:** Consistency ensures that similar elements behave in similar ways throughout the interface. This includes consistent use of terminology, layout, colors, fonts, and navigation patterns.

Standards refer to established conventions or best practices within the industry. Standards can encompass various aspects of design, such as button placement, menu structures, form layouts, and interaction patterns.

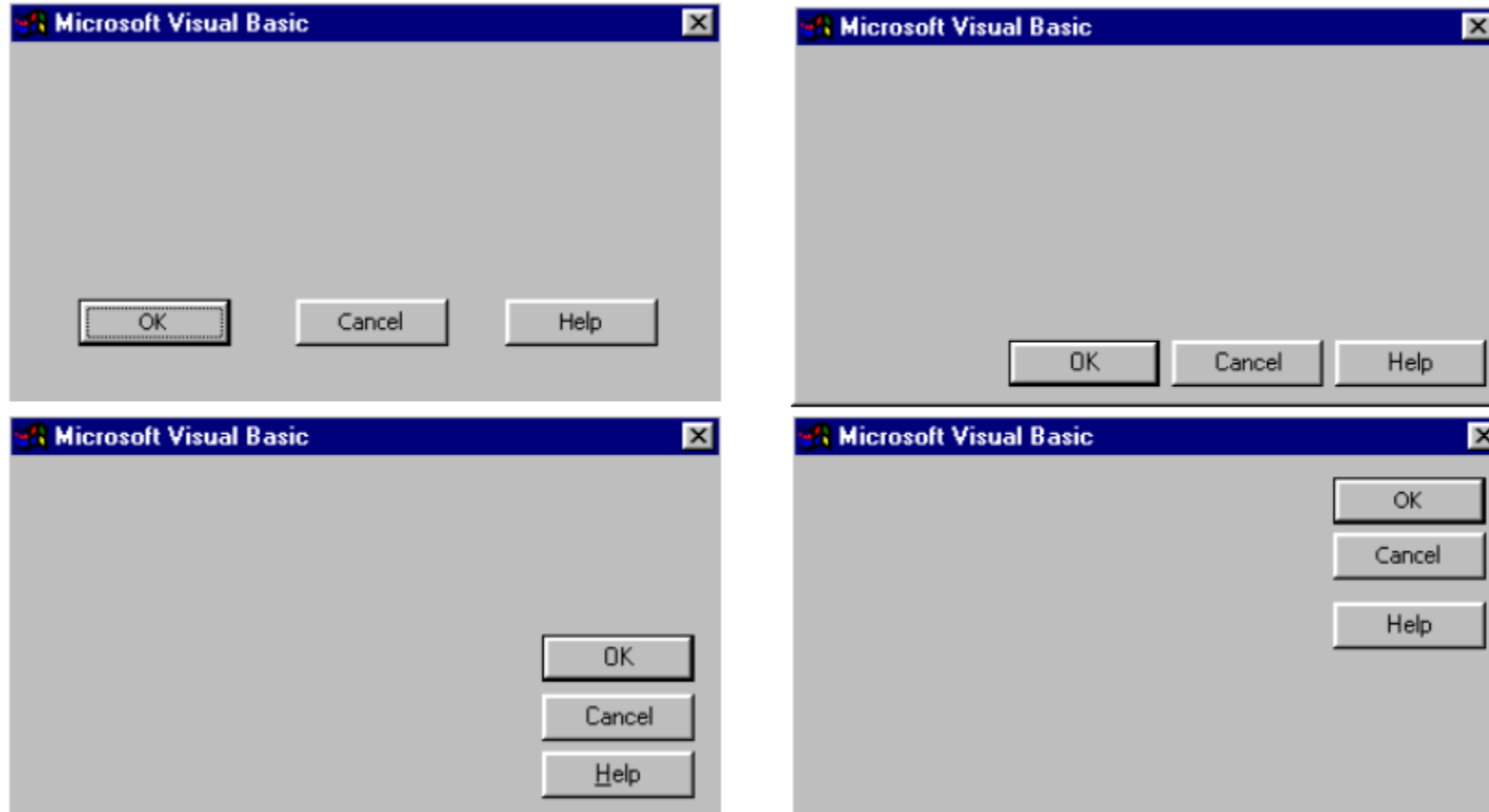


Benefits of consistency and standards:

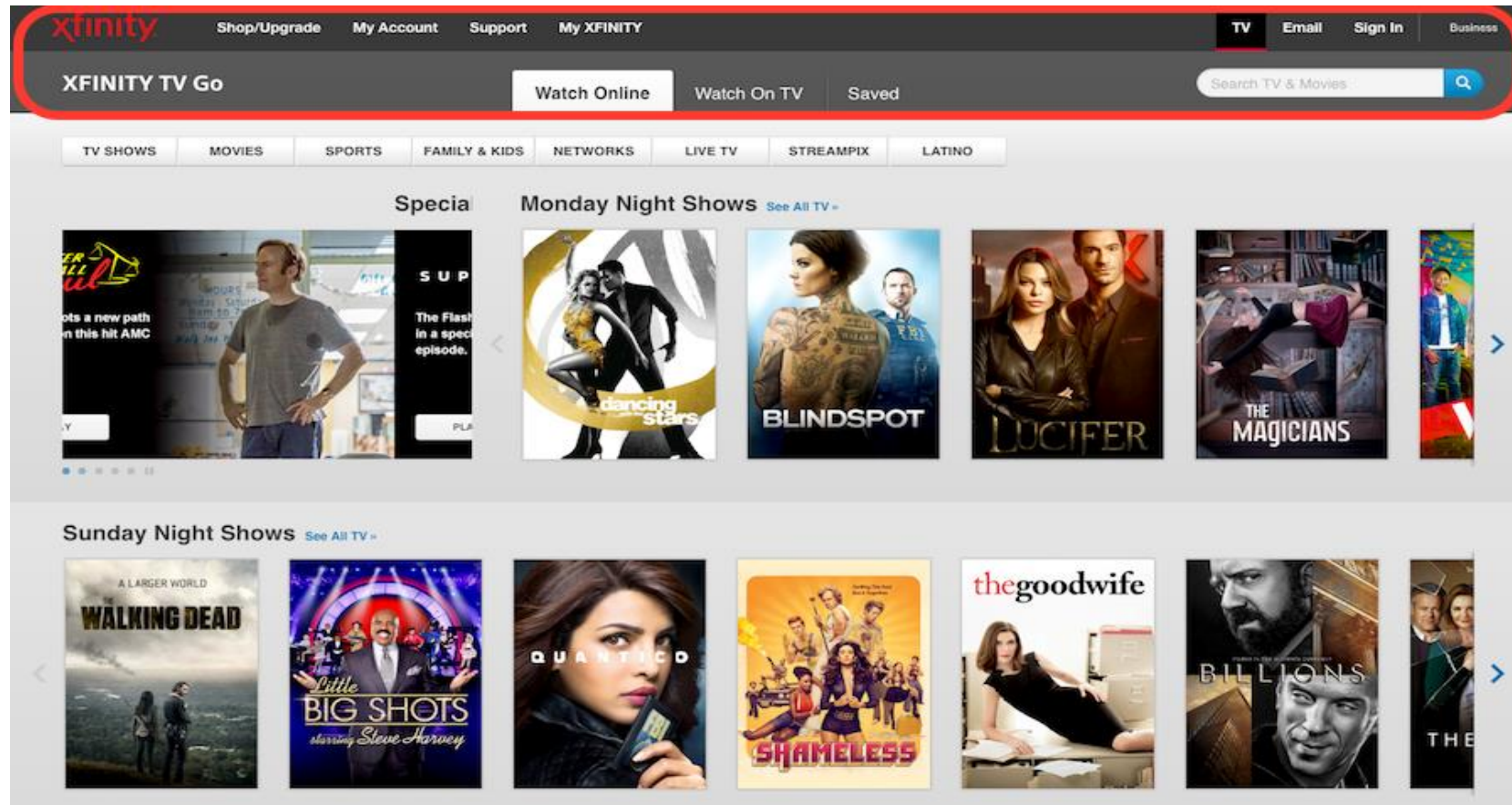
- 1) **Reduce learning:** Users can quickly learn how to navigate an interface if it follows familiar standards and maintains consistency in its design elements.
- 2) **Eliminate confusion:** Consistent design patterns and adherence to standards allow users to focus on their tasks without having to learn new ways of interacting with each element.



Consistency and standards



Consistency and standards



Consistency and standards

The screenshot displays the Xfinity website homepage with a red navigation bar at the top containing links for 'My XFINITY', 'Shop/Upgrade', 'Support', and 'My Account'. On the right side of the bar are icons for a monitor, email, a menu, and a 'Sign In' button. Below the navigation bar, the Xfinity logo is on the left, followed by a search bar labeled 'Search the Web' with a 'SEARCH' button. To the right of the search bar is the weather '53°' with a sun icon and a link to 'Make XFINITY My Homepage'. The main content area features several large tiles: a 'TELEVISION' tile for 'DWTS' Week 2, a 'Watch Live Soccer' tile for a 'World Cup Qualifier' on ESPN2, a 'YOU COULD WIN \$10K' sweepstakes tile, and an 'mbna' credit card advertisement for 'Zero interest'. Below these tiles is a horizontal menu with categories: 'Top', 'News', 'Sports', 'Finance', 'TV', 'Entertainment', 'Life', 'Games', and 'Videos'. The 'News' section is active, showing a headline about Obama's fight against opioid addiction. The 'Sports' section shows headlines about Hield and Valentine's AP All-America team selection and NFL concussion stories.

My XFINITY Shop/Upgrade Support My Account

Sign In

xfinity

Search the Web SEARCH

enhanced by Google

53°

Make XFINITY My Homepage

YOU COULD WIN \$10K
Enter the XFINITY Watchathon Sweepstakes

Zero interest
You could pay off your credit card balance sooner with 0% interest* on balance transfers* for 12 months.

Apply Now mbna

Ad Info Ad Feedback

Top News Sports Finance TV Entertainment Life Games Videos

TOP NEWS

Obama Ramps Up Fight Against Opioid Addiction

Ted Cruz Nabs Key Endorsement

Jilted Lotto Winner Files for Bankruptcy

Home with Alleged 'Watcher' Up for Sale

Feds Charge Principals in Kickback Scheme

SPORTS

Hield, Valentine Headline AP All-America Team

NFL Threatens Times over Concussion Story

Owners Face Backlash over CTE Remarks

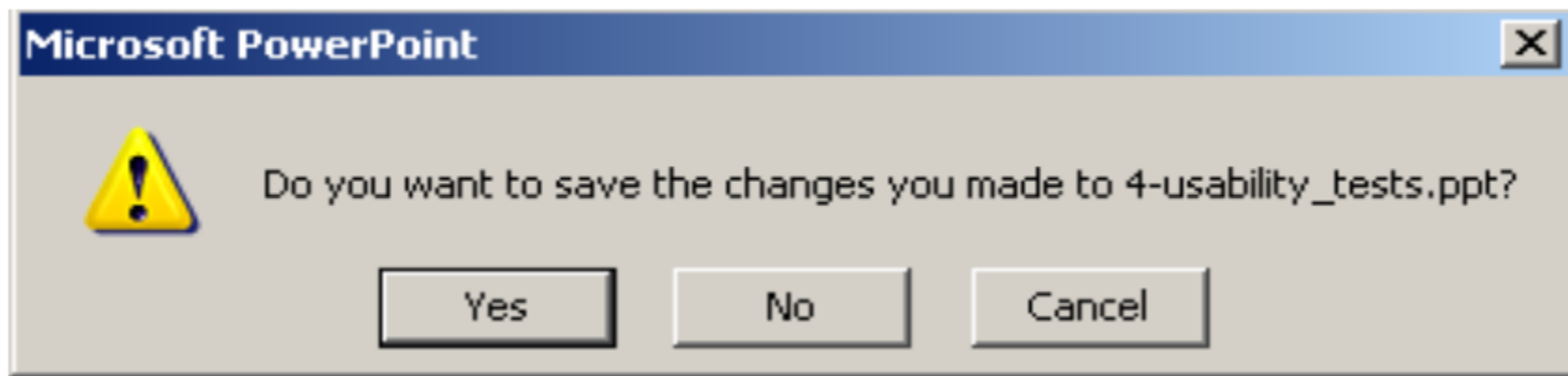
LeBron Partners with Major Golf Tournament

Watch ESPN for News and Highlights

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5) **Error prevention:** aimed at minimizing the occurrence of mistakes and providing users with mechanisms to recover from errors when they do occur. These principles help ensure a smoother and more error-tolerant user experience

For example in case of closing unsaved document



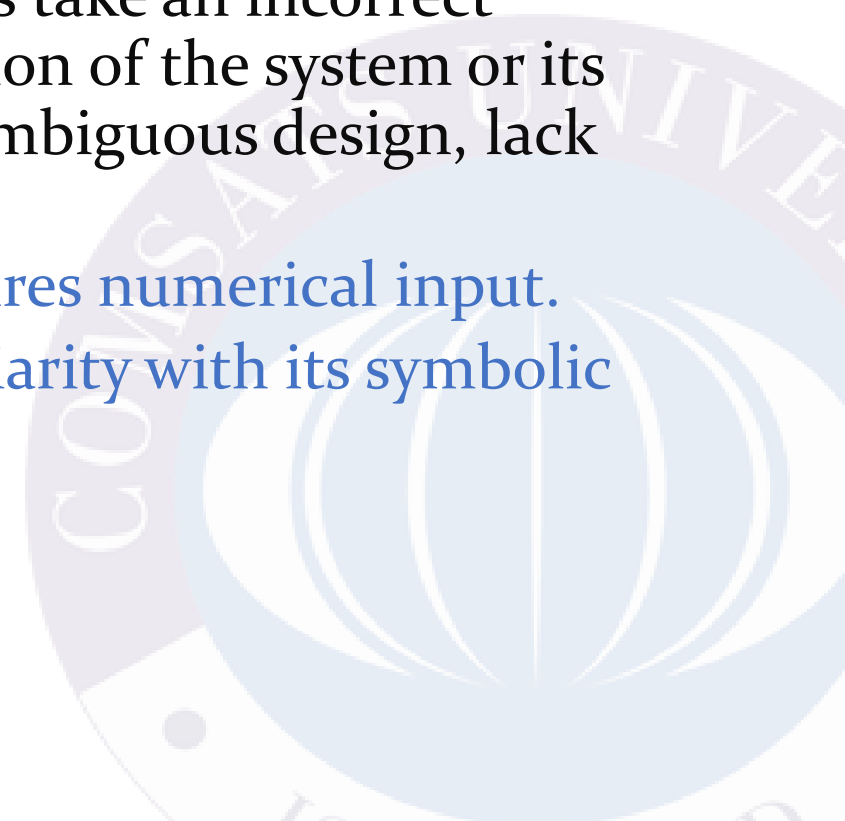
Slips are unconscious errors caused by inattention

Slips occur when users perform an action incorrectly despite knowing the correct action to take.

- Pressing the wrong key on a keyboard while typing quickly.
- Forgetting to attach a file to an email despite intending to do so.

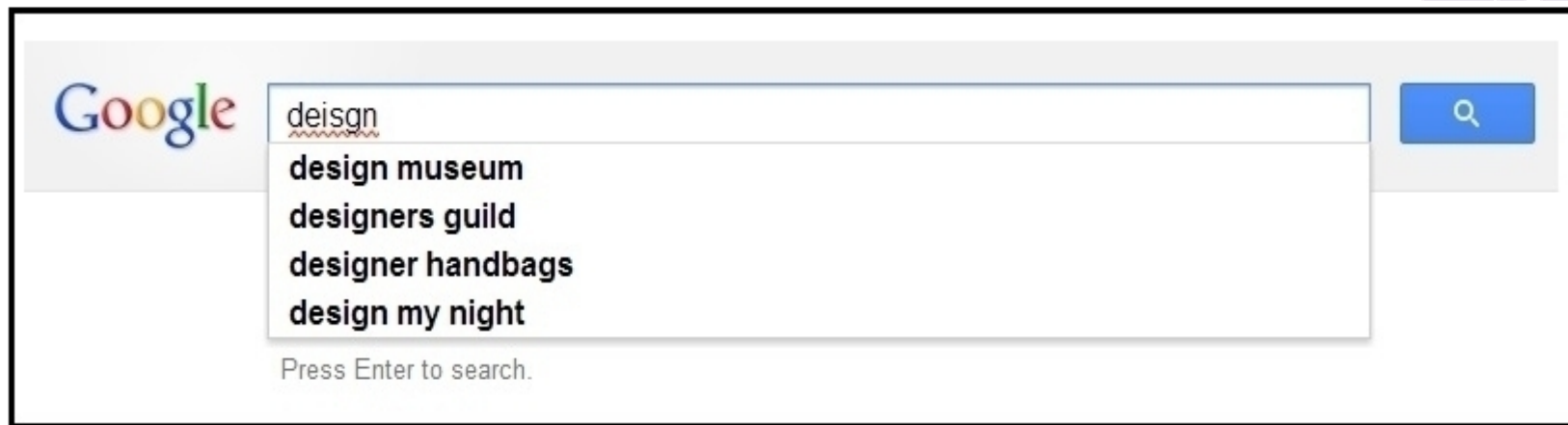
Mistakes: Mistakes are errors that occur when users take an incorrect action due to a misunderstanding or misinterpretation of the system or its interface. Mistakes can result from factors such as ambiguous design, lack of feedback, inadequate user training.

- Entering alphabetic characters in a field that requires numerical input.
- Misinterpreting an icon's meaning due to unfamiliarity with its symbolic representation.



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Google search helps users by offering suggestions based on their input. As you can see, even though the user spelled design incorrectly the suggestions provide them with a means of instantly correcting their mistake, by selecting one of the options below.

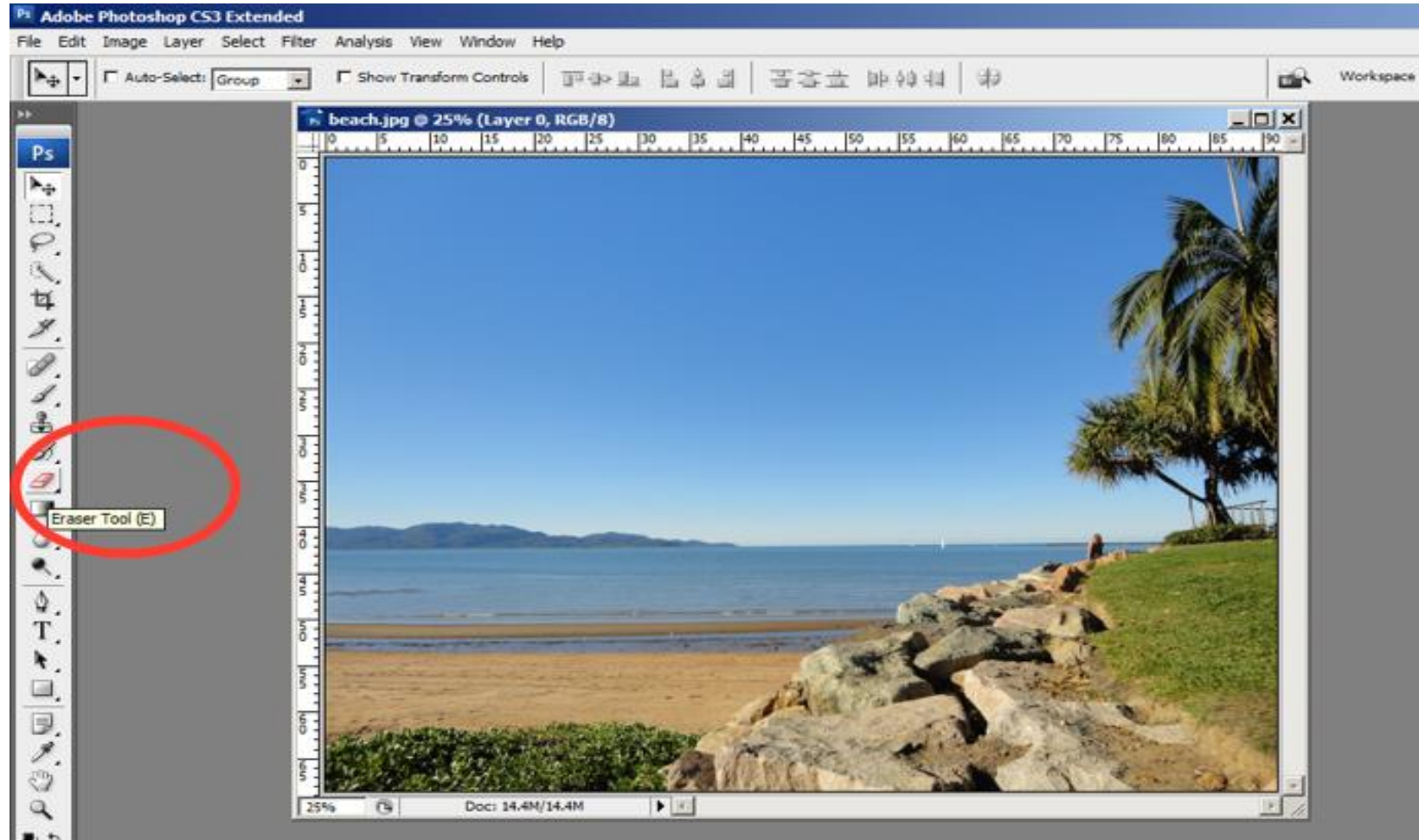


Nielsen and Molich's 10 User Interface Design Guidelines...

- To prevent users from making errors, Photoshop provides a brief description or label of the tools when a user floats over it to help make sure users are using the proper tool for the task at hand.



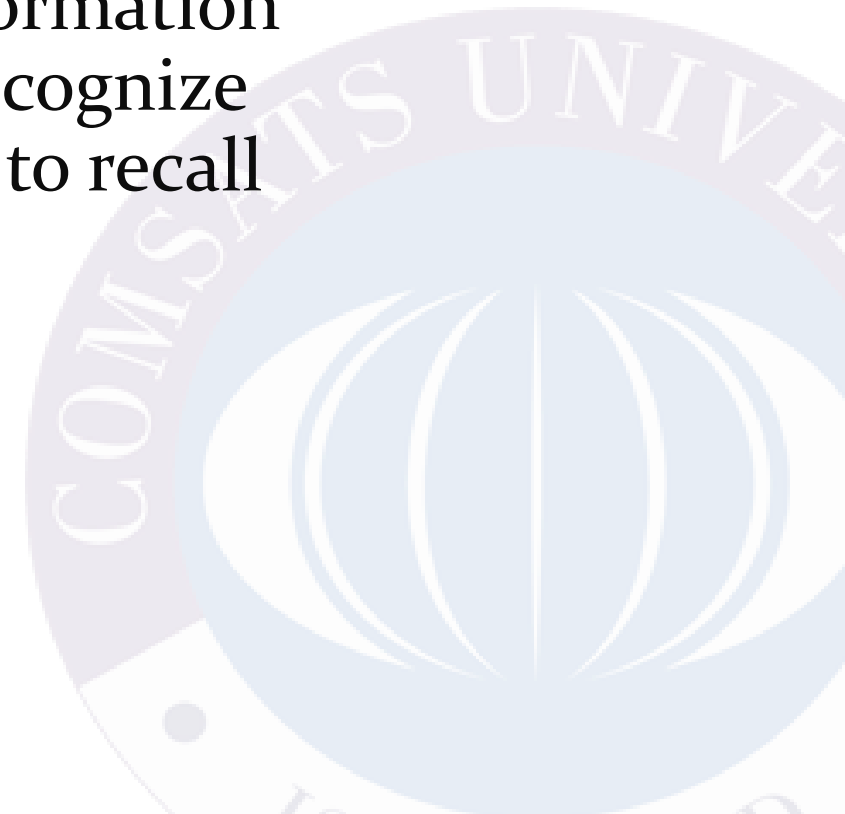
Nielsen and Molich's 10 User Interface Design Guidelines...



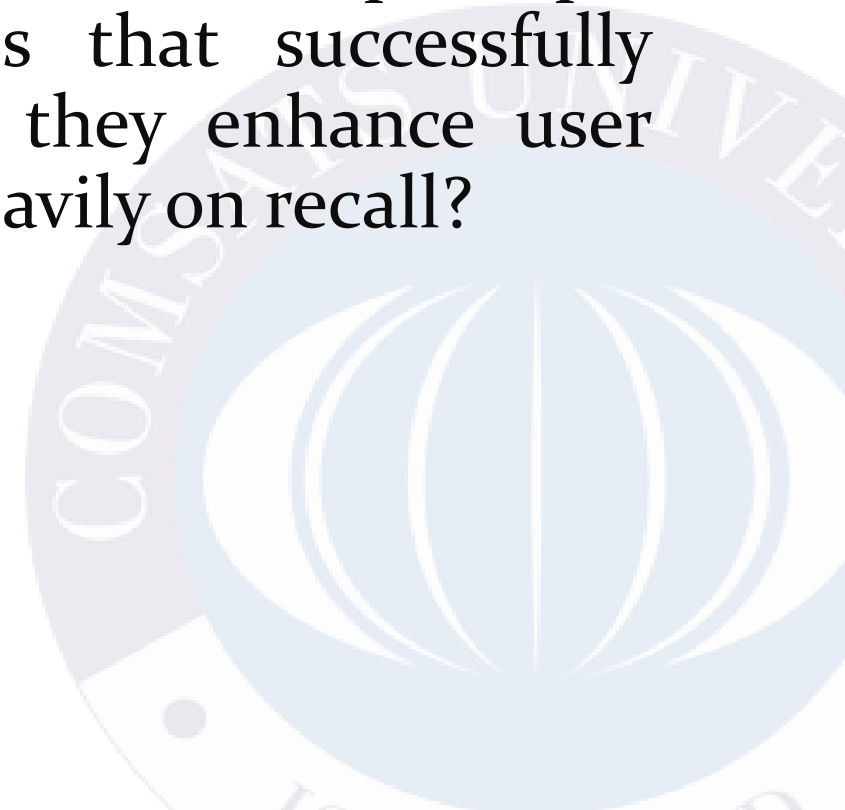
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6) **Recognition rather than recall:**

This principle highlights the importance of designing interfaces in a way that minimizes the cognitive load on users by presenting information and options in a manner that is easy to recognize and understand, without requiring users to recall specific details from memory.

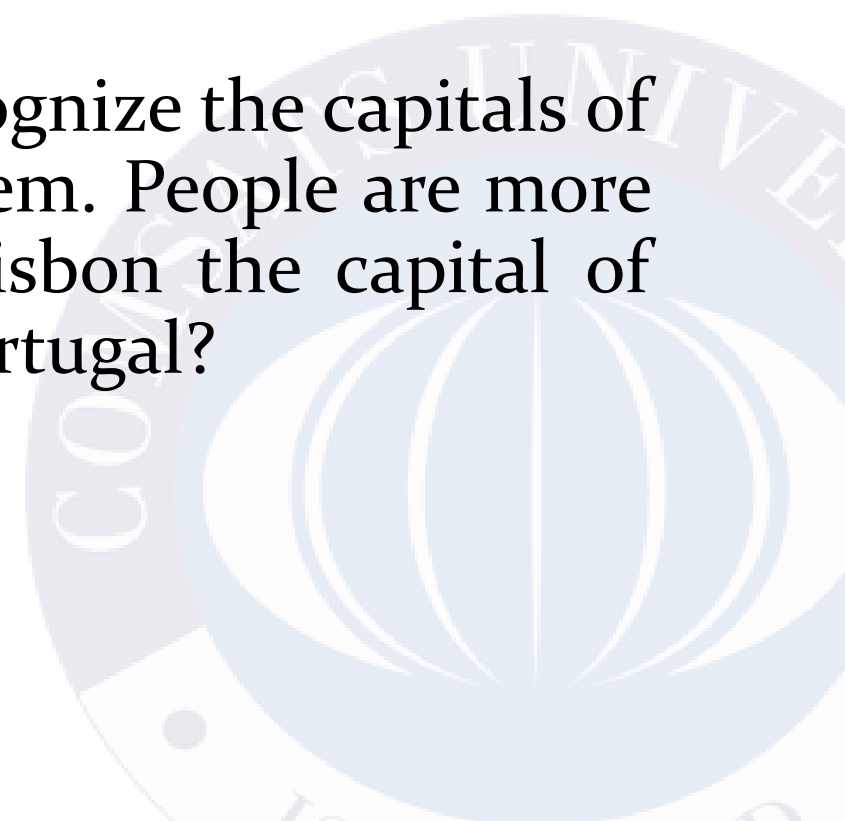


- How does the principle of 'Recognition rather than recall' contribute to the usability of user interfaces, and what are some strategies that designers can employ to adhere to this principle effectively? Provide examples of interfaces that successfully implement this principle and discuss how they enhance user experience compared to interfaces that rely heavily on recall?



Importance

- Interfaces that promote recognition reduce the amount of cognitive effort required from users.
- For example, It's easier for most people to recognize the capitals of countries, instead of having to remember them. People are more likely to correctly answer the question Is Lisbon the capital of Portugal? rather than What's the capital of Portugal?

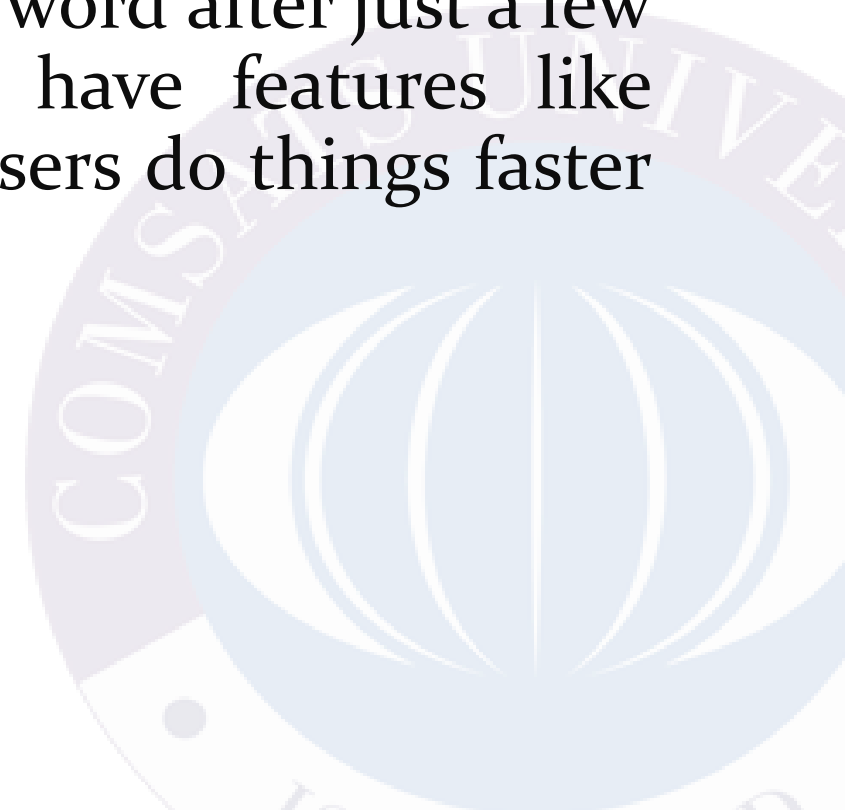


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7) Flexibility and efficiency of use:

Flexibility: Think of flexibility like a tool belt. Everyone has different tools they like to use depending on the job they're doing. In interface design, flexibility means giving users options to customize their experience. For example, just like you might adjust the seat in a car to fit your height, flexible interfaces let users change things like font size, color schemes, or even where buttons are placed. This makes the interface more comfortable and easier to use for different people.

- **Efficiency of Use:** Efficiency is like taking a shortcut to get where you want to go faster. In interface design, efficiency of use means making it quicker and easier for users to do things they need to do. For example, imagine you're texting on your phone and you want to type "hello." Instead of tapping each letter individually, you might use predictive text that guesses the word after just a few taps. That's efficiency! Efficient interfaces have features like shortcuts or quick access buttons that help users do things faster and with less effort.



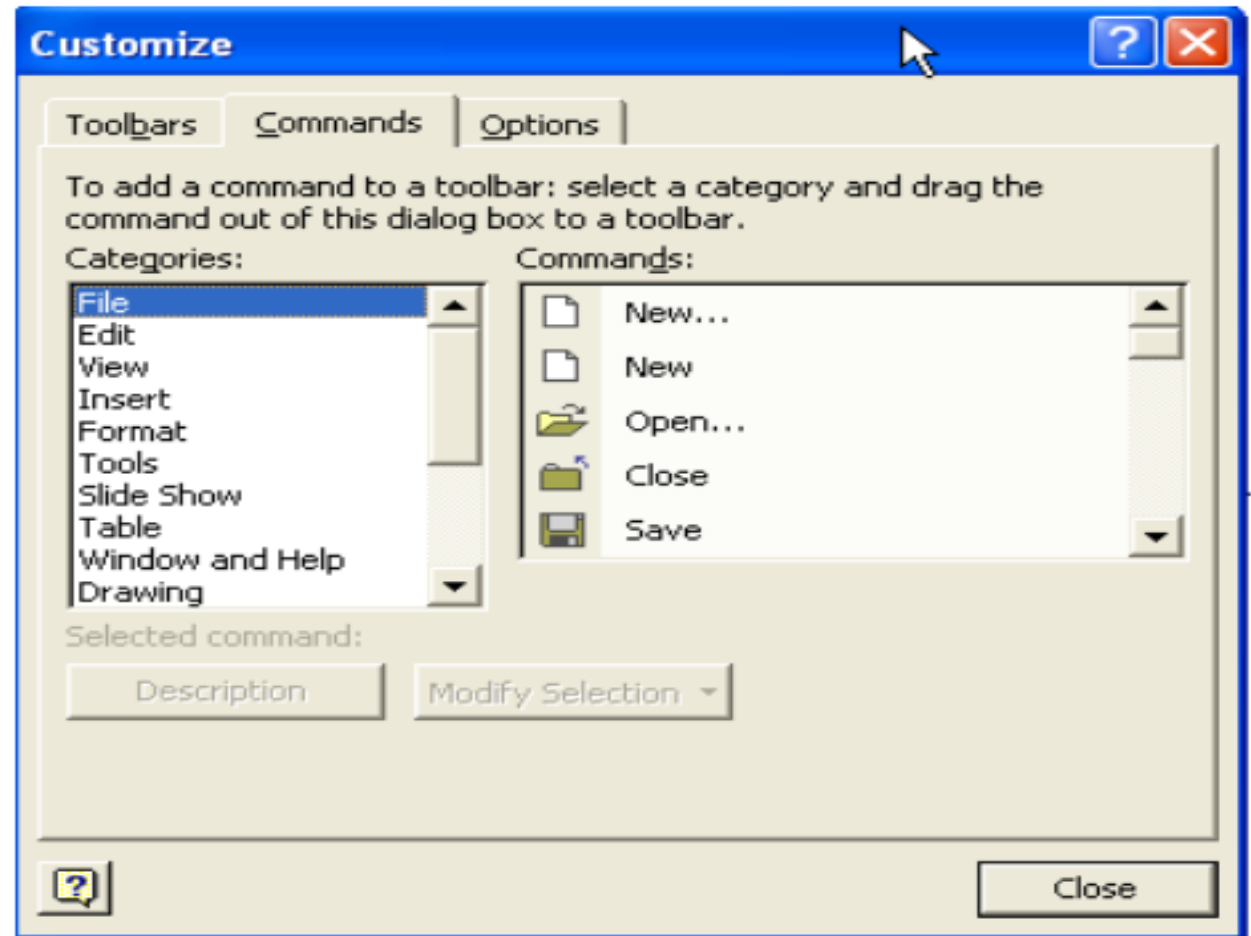
Flexibility and efficiency of use:

- flexibility is about making the interface adaptable to different users, like adjusting a tool belt, while efficiency of use is about making it quick and easy to get things done, like finding a shortcut to your destination



Nielsen and Molich's 10 User Interface Design Guidelines...

- Provide personalization by tailoring content and functionality for individual users.
- Allow for customization, so users can make selections about how they want the product to work.



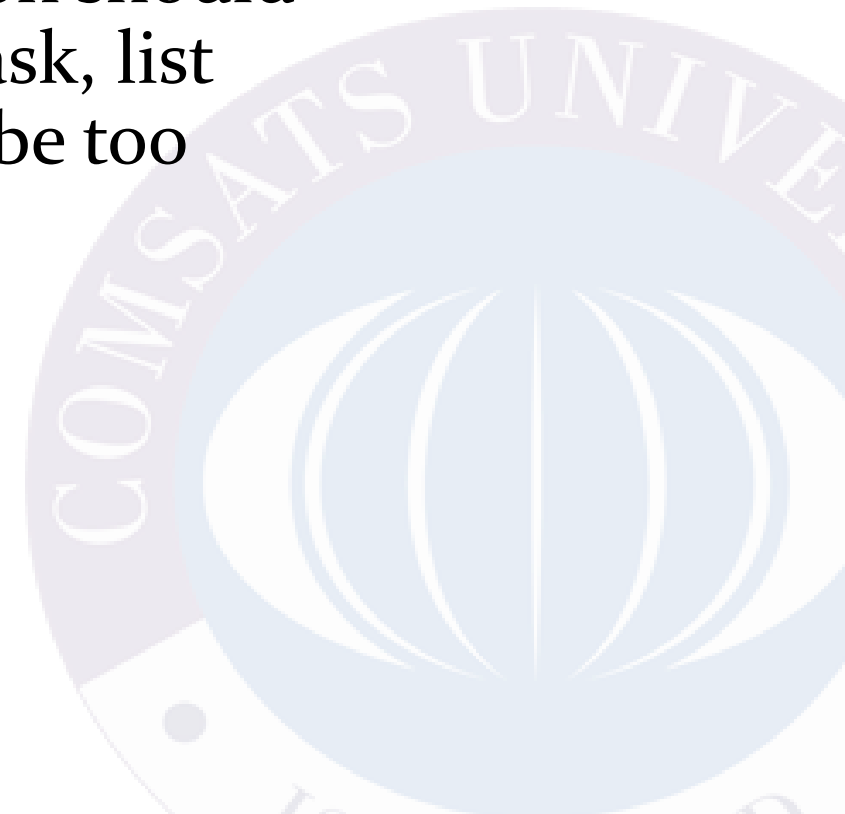
Nielsen and Molich's 10 User Interface Design Guidelines...

Provide accelerators like keyboard shortcuts and touch gestures.

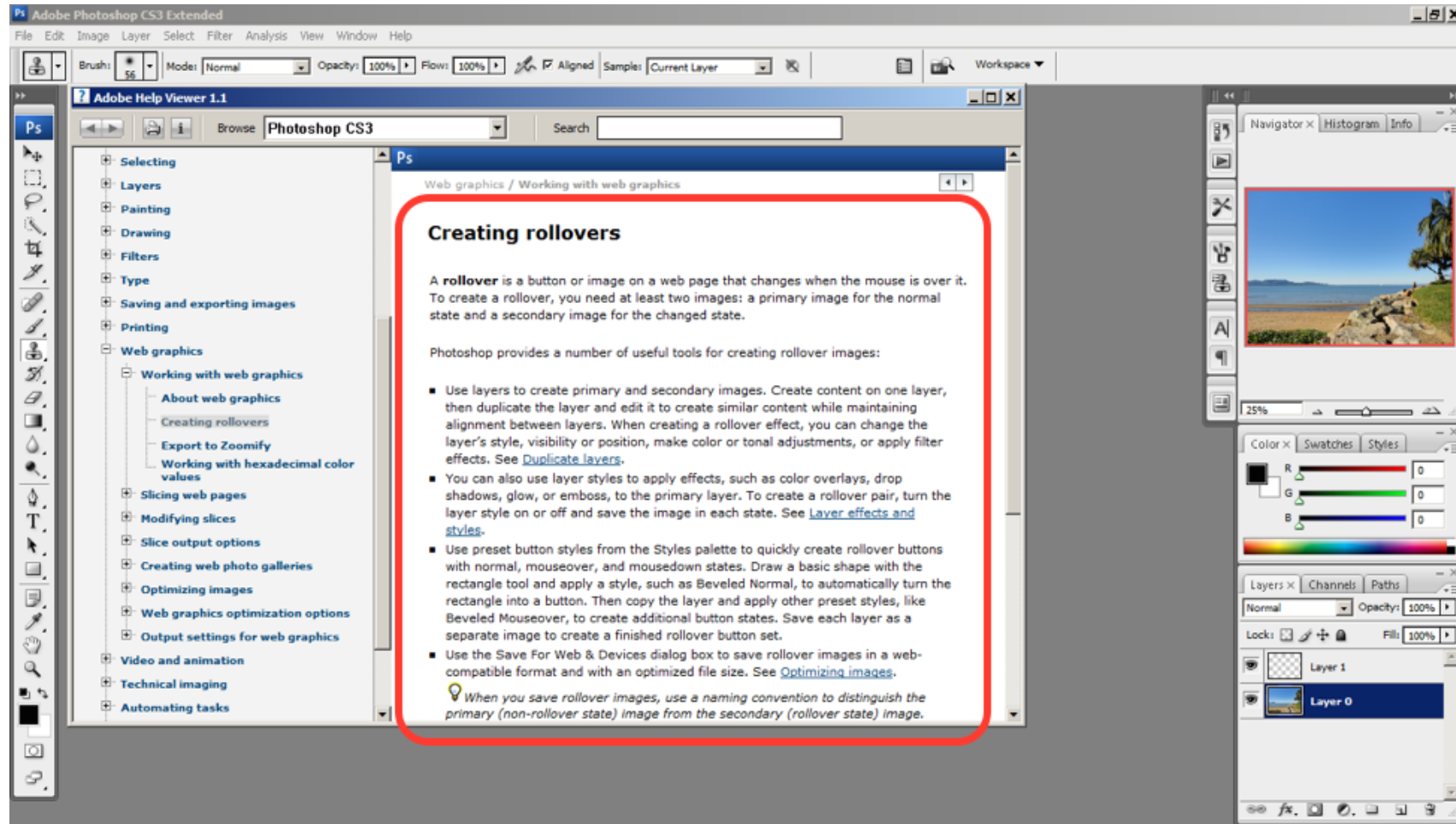
1	Cmd-C	Copy files
2	Cmd-V	Paste files
3	Option-Drag	Copy files to new location
4	Cmd-Drag	Move and auto-align icons
5	Cmd-Delete	Delete
6	Cmd-Option-Drag	Make alias
7	Cmd-Shift-Delete	Empty trash
8	Cmd-Shift-Option-Delete	Empty trash without warning
9	Tab	Select next field
10	Shift-Tab	Select previous field
11	Return	Perform default action
12	Escape	Close dialog box
13	Page Up	Scroll up list
14	Up Arrow	Select item above
15	Page Down	Scroll down list
16	Down Arrow	Select item below
17	Cmd-Shift-G	Open 'Go to Folder' dialog
18	Cmd-Period[.]	Close dialog box

Nielsen and Molich's 10 User Interface Design Guidelines...

8) **Help and documentation:** Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.



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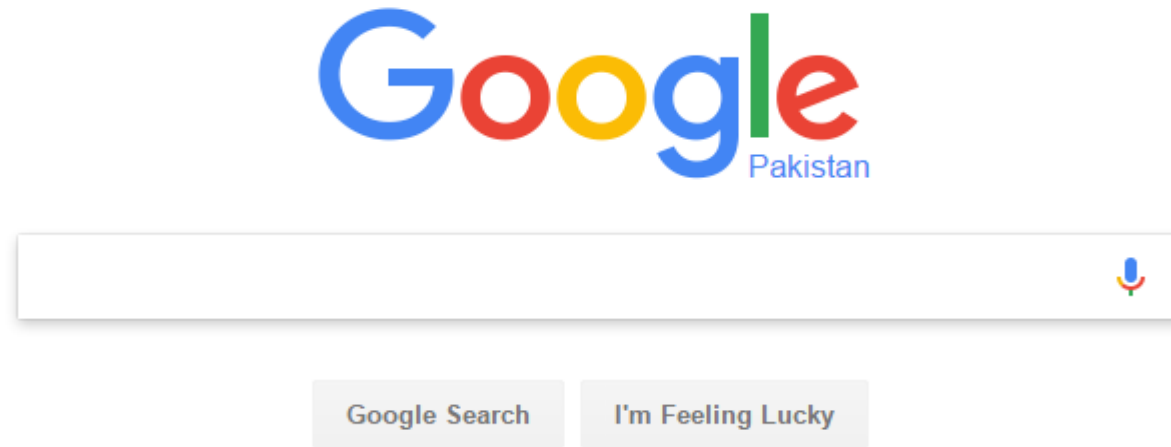


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9) **Aesthetic and minimalist design:** Dialogues should not contain information which is irrelevant or rarely.



Nielsen and Molich's 10 User Interface Design Guidelines...



Nielsen and Molich's 10 User Interface Design Guidelines...

10) **Help users recognize, diagnose, and recover from errors:** Error message should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

The image shows two side-by-side screenshots of web interfaces. The left screenshot is the Amazon sign-in page. It features the Amazon logo at the top. Below it, a red-bordered box contains a warning icon and the text "There was a problem" followed by "Your password is incorrect". Below this is the "Sign-In" section with a phone number field (containing "+919712729389"), a password field, and a "Sign-In" button. There are links for "Forgot your password?" and "Keep me signed in. Details". The right screenshot is a rental furniture form. It has a header "Tell us more about yourself and your rental furniture needs here." followed by several input fields: "Name" (split into "First Name" with "Harsh" and "Last Name" with "Gorasia"), "Phone Number", "Email" (containing "sdsdsd.com" with a red error message "Enter a valid email address. (eg: yourname@domain.com)"), "Tell us about your Profession" (a dropdown menu with "Salaried" selected), "Choose the city where you want to rent furniture" (a dropdown menu with "-Select-" selected), and "Where do you stay" (a dropdown menu with "-Select-" selected). There is also a text area for "Add any special requests/comments here".

amazon

! There was a problem
Your password is incorrect

Sign-In

+919712729389 [Change](#)

Password [Forgot your password?](#)

Sign-In

☐ Keep me signed in. [Details](#)

[Conditions of Use](#) [Privacy Notice](#) [Help](#)

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Tell us more about yourself and your rental furniture needs here.

Name *
First Name Last Name

Phone Number *

Email *
Enter a valid email address. (eg: yourname@domain.com)

Tell us about your Profession *

Choose the city where you want to rent furniture *

Where do you stay? *

Add any special requests/comments here

Nielsen and Molich's 10 User Interface Design Guidelines...



Evaluating UIs' for Nielsen and Molich's Design Guidelines...

Recognizing Nielsen and Molich's User Interface Design Guidelines

Step 1: Choose a website or an app.

Step 2: Put it to the test! Look at whether or not it follows Nielsen and Molich's 10 rules of thumb.

Rule of Thumb	Is this rule being applied? How so?	Is this rule violated? How so?
1. Visibility of system status		
2. Match between system and the real world		
3. User control and freedom		
4. Consistency and standards		
5. Error prevention		
6. Recognition rather than recall		
7. Flexibility and efficiency of use		
8. Aesthetic and minimalist design		
9. Help users recognize, diagnose and recover from errors		
10. Help and documentation		

